

Problem 10 Solution

Shoshaunah Jacob, 11/13/17

Given a force field $\vec{F}$, find the associated potential energy function E_p .

Motivation

All conservative force fields have an associated potential energy function, which depends only on location. Knowing this potential energy function means you can calculate the change in energy when the particle or rigid body moves from one location to another without knowing the exact path or having to do a path integral of the force every time.

Methodology

Although all the force fields given are conservative, you will at some point come across a non-conservative force field (or Professor Ruina could have been sneaky and given us an unsolvable problem). One way to check if a force field is conservative is to take $E_p = - \int_{\vec{r}_0}^{\vec{r}} \vec{F} \cdot d\vec{r}$ and check that the integral is path-independent before solving it. A second way is to use $-\nabla E_p(\vec{r}) = \vec{F}$ and back-checking the solution for E_p for validity.

Math facts used in this problem:

$$d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k} = dr\hat{e}_r + r d\theta\hat{e}_\theta + dz\hat{k}$$
$$\nabla E_p = \frac{\partial E_p}{\partial x}\hat{i} + \frac{\partial E_p}{\partial y}\hat{j} + \frac{\partial E_p}{\partial z}\hat{k} = \frac{\partial E_p}{\partial r}\hat{e}_r + \frac{1}{r}\frac{\partial E_p}{\partial \theta}\hat{e}_\theta + \frac{\partial E_p}{\partial z}\hat{k}$$

A relevant proof

The following equation will be used several times throughout these solutions.

$$\vec{A} \cdot d\vec{A} = |\vec{A}| d|\vec{A}|$$

A short proof of this equation is below:

$$\vec{A} \cdot \vec{A} = |\vec{A}|^2$$

$$\frac{d(\vec{A} \cdot \vec{A})}{dt} = \frac{d|\vec{A}|^2}{dt}$$

$$2\vec{A} \frac{d\vec{A}}{dt} = 2|\vec{A}| \frac{d|\vec{A}|}{dt}$$

$$\vec{A} \frac{d\vec{A}}{dt} = |\vec{A}| \frac{d|\vec{A}|}{dt}$$

$$\vec{A} \cdot d\vec{A} = |\vec{A}| d|\vec{A}|$$

(a)

Given $\vec{F} = -mg\hat{j}$ find E_p . Chosen method: $E_p = -\int_{\vec{r}_0}^{\vec{r}} \vec{F} \cdot d\vec{r}$. This force is like near-earth gravity.



Sketch of force

$$E_p = -\int_{\vec{r}_0}^{\vec{r}} -mg\hat{j} \cdot d\vec{r} = mg \int_{\vec{r}_0}^{\vec{r}} \hat{j} \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) = mg \int_{y_0}^y dy$$

This integral is path-independent.

$$mg \int_{y_0}^y dy = mg(y - y_0)$$

Since y_0 is an arbitrary constant (from the arbitrary starting vector), choose $y_0 = 0$ to find E_p .

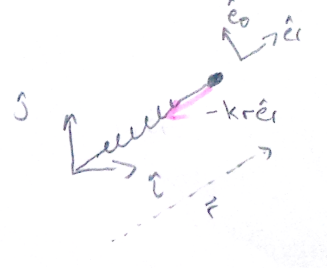
$$E_p = mgy$$

(b)

Given $\vec{F} = -kr\hat{e}_r = -k\vec{r}$ find E_p .

Chosen method: $-\nabla E_p(\vec{r}) = \vec{F}$.

This force is like a zero-rest-length spring.



Sketch of force

$$-\frac{\partial E_p}{\partial r} = \vec{F} \cdot \hat{e}_r = -kr \quad -\frac{1}{r} \frac{\partial E_p}{\partial \theta} = \vec{F} \cdot \hat{e}_\theta = 0 \quad -\frac{\partial E_p}{\partial z} = \vec{F} \cdot \hat{k} = 0$$

From this, we can integrate both sides of $-\frac{\partial E_p}{\partial r} = -kr$ to find E_p

$$E_p = - \int -kr dr = \frac{k}{2} r^2 + g(\theta, z)$$

It is important to include $g(\theta, z)$ because taking the partial derivative with respect to r would have gotten rid of all terms that only depended on θ or z , as well as constants. Now we need to find what $g(\theta, z)$ is by taking the partial derivatives of E_p .

$$\frac{1}{r} \frac{\partial E_p}{\partial \theta} = \frac{1}{r} \left(0 + \frac{\partial g}{\partial \theta} \right) = 0$$

$$\text{Therefore } \frac{\partial g}{\partial \theta} = 0$$

From this, we know that g is only a function of z , $g(z)$.

$$\frac{\partial E_p}{\partial z} = \frac{dg}{dz} = 0$$

From this we know $g(z) = C$, so that $E_p = \frac{k}{2} r^2 + C$. Choose $C = 0$ and take the gradient of E_p to confirm that it is correct.

$$-\nabla E_p = -\frac{\partial E_p}{\partial r} \hat{e}_r - \frac{1}{r} \frac{\partial E_p}{\partial \theta} \hat{e}_\theta - \frac{\partial E_p}{\partial z} \hat{k} = -(kr) \hat{e}_r - \frac{1}{r} (0) \hat{e}_\theta - (0) \hat{k} = -kr \hat{e}_r = \vec{F}$$

Since the negative gradient of E_p is our original $\vec{F}$, we know that

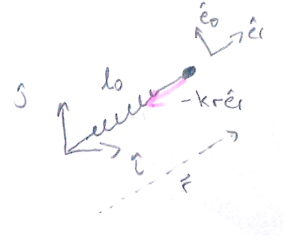
$$E_p = \frac{k}{2} r^2$$

(c)

Given $\vec{F} = -k(r - l_0)\hat{e}_r$ find E_p .

Chosen method: $E_p = -\int_{\vec{r}_0}^{\vec{r}} \vec{F} \cdot d\vec{r}$.

This force is like a spring with rest length l_0 .



Sketch of force

I have changed the original force vector to use r instead of l for consistency; both represent the distance from the origin. l_0 is the rest length of the spring.

$$E_p = -\int_{\vec{r}_0}^{\vec{r}} -k(r - l_0)\hat{e}_r \cdot d\vec{r} = k \int_{\vec{r}_0}^{\vec{r}} (r - l_0)\hat{e}_r \cdot (dr\hat{e}_r + r d\theta\hat{e}_\theta + dz\hat{k}) = k \int_{r_0}^r (r - l_0)dr$$

This integral is path independent.

$$k \int_{r_0}^r (r - l_0)dr = \frac{k}{2}(r^2 - 2l_0r) \Big|_{r_0}^r = \frac{k}{2}(r^2 - 2l_0(r - r_0) - r_0^2)$$

Choose $r_0 = l_0$. While this might seem like an odd choice, it simplifies the equation nicely:

$$\frac{k}{2}(r^2 - 2l_0(r - l_0) - l_0^2) = \frac{k}{2}(r^2 - 2l_0r + 2l_0^2 - l_0^2) = \frac{k}{2}(r^2 - 2l_0r + l_0^2)$$

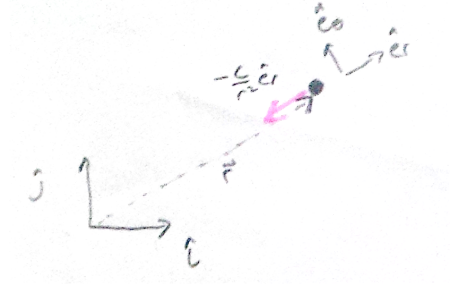
$$E_p = \frac{k}{2}(r - l_0)^2$$

(d)

Given $\vec{F} = \frac{-C}{r^3} \vec{r} = \frac{-C}{r^2} \hat{e}_r$ find E_p .

Chosen method: $-\nabla E_p(\vec{r}) = \vec{F}$.

This force is like inverse-square gravity towards something at the origin.



Sketch of force

$$-\frac{\partial E_p}{\partial r} = \vec{F} \cdot \hat{e}_r = \frac{-C}{r^2} \quad -\frac{1}{r} \frac{\partial E_p}{\partial \theta} = \vec{F} \cdot \hat{e}_\theta = 0 \quad -\frac{\partial E_p}{\partial z} = \vec{F} \cdot \hat{k} = 0$$

From this, we can integrate both sides of $-\frac{\partial E_p}{\partial r} = \frac{-C}{r^2}$ to find E_p

$$E_p = - \int \frac{-C}{r^2} dr = \frac{-C}{r} + g(\theta, z)$$

It is important to include $g(\theta, z)$ because taking the partial derivative with respect to r would have gotten rid of all terms that only depended on θ or z , as well as constants. Now we need to find what $g(\theta, z)$ is by taking the partial derivatives of E_p .

$$\frac{1}{r} \frac{\partial E_p}{\partial \theta} = \frac{1}{r} (0 + \frac{\partial g}{\partial \theta}) = 0$$

$$\text{Therefore } \frac{\partial g}{\partial \theta} = 0$$

From this, we know that g is only a function of z , $g(z)$.

$$\frac{\partial E_p}{\partial z} = \frac{dg}{dz} = 0$$

From this we know $g(z) = C_0$, so that $E_p = \frac{-C}{r} + C_0$. Choose $C_0 = 0$ and take the gradient of E_p to confirm that it is correct.

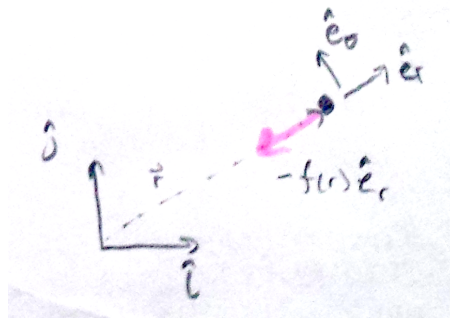
$$-\nabla E_p = -\frac{\partial E_p}{\partial r} \hat{e}_r - \frac{1}{r} \frac{\partial E_p}{\partial \theta} \hat{e}_\theta - \frac{\partial E_p}{\partial z} \hat{k} = -\left(\frac{C}{r^2}\right) \hat{e}_r - \frac{1}{r} (0) \hat{e}_\theta - (0) \hat{k} = \frac{-C}{r^2} \hat{e}_r = \vec{F}$$

Since the negative gradient of E_p is our original $\vec{F}$, we know that

$$E_p = \frac{-C}{r}$$

(e)

Given $\vec{F} = -f(r)\hat{e}_r$ find E_p . Chosen method: $E_p = -\int_{\vec{r}_0}^{\vec{r}} \vec{F} \cdot d\vec{r}$. This force is an unknown central force.



Sketch of force

$$E_p = -\int_{\vec{r}_0}^{\vec{r}} -f(r)\hat{e}_r \cdot d\vec{r} = \int_{\vec{r}_0}^{\vec{r}} f(r)\hat{e}_r \cdot (dr\hat{e}_r + r d\theta\hat{e}_\theta + dz\hat{k}) = \int_{r_0}^r f(r)dr$$

This integral is path independent.

Since we don't know the exact form of f , we can't simplify this integral further. However, it is important to choose a value of r_0 that will not cause the integral to become divergent. For instance, if $f(r) = \frac{1}{r}$, and we chose $r_0 = 0$, then $f(r_0)$ would be undefined and we couldn't solve it. Therefore,

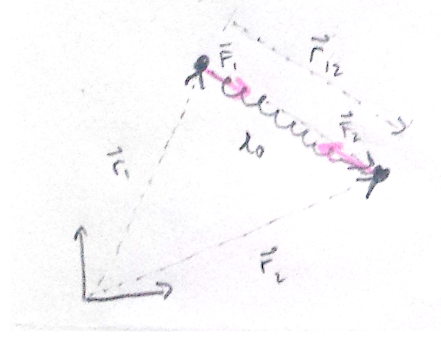
$$E_p = \int_{r_0}^r f(r)dr \text{ with } r_0 \text{ chosen so the integral is not divergent}$$

Also see part (j) for a similar unknown force in the $\hat{i}$ direction.

(f)

Given $\vec{F}_1 = -\vec{F}_2 = k(|\vec{r}_{12}| - l_0) \frac{\vec{r}_{12}}{|\vec{r}_{12}|}$ find E_p .

Chosen method: $E_p = - \int_{\vec{r}_0}^{\vec{r}} \vec{F} \cdot d\vec{r}$. This force is like a spring with rest length l_0 between two objects.



Sketch of force

In this case, because we have two particles and therefore two positions that change, we need to sum the two changes in potential energy. There are also two initial and two final positions.

$$E_p = - \int_{\vec{r}_{10}, \vec{r}_{20}}^{\vec{r}_1, \vec{r}_2} \vec{F}_1 \cdot d\vec{r}_1 + \vec{F}_2 \cdot d\vec{r}_2 = \int_{\vec{r}_{10}, \vec{r}_{20}}^{\vec{r}_1, \vec{r}_2} -\vec{F}_1 \cdot d\vec{r}_1 + \vec{F}_1 \cdot d\vec{r}_2 = \int_{\vec{r}_{10}, \vec{r}_{20}}^{\vec{r}_1, \vec{r}_2} \vec{F}_1 \cdot (d\vec{r}_2 - d\vec{r}_1)$$

Because $\vec{r}_1$ and $\vec{r}_2$ are two independent vectors, $d\vec{r}_2 - d\vec{r}_1 = d(\vec{r}_2 - \vec{r}_1) = d\vec{r}_{12}$. Therefore

$$\int_{\vec{r}_{10}, \vec{r}_{20}}^{\vec{r}_1, \vec{r}_2} \vec{F}_1 \cdot (d\vec{r}_2 - d\vec{r}_1) = \int_{\vec{r}_{10}, \vec{r}_{20}}^{\vec{r}_1, \vec{r}_2} \vec{F}_1 \cdot d\vec{r}_{12} = \int_{\vec{r}_{120}}^{\vec{r}_{12}} k(|\vec{r}_{12}| - l_0) \frac{\vec{r}_{12}}{|\vec{r}_{12}|} \cdot d\vec{r}_{12}$$

Using the fact that $\vec{A} \cdot d\vec{A} = |\vec{A}| d|\vec{A}|$, and defining $l_{12} = |\vec{r}_{12}|$, this is equal to

$$\int_{l_{120}}^{l_{12}} k(l_{12} - l_0)(1)dl_{12} = \frac{k}{2}(l_{12}^2 - 2l_0l_{12}) \Big|_{l_{120}}^{l_{12}} = \frac{k}{2}(l_{12}^2 - 2l_0(l_{12} - l_{120}) - l_{120}^2)$$

Choose $l_{120} = l_0$. Then

$$\frac{k}{2}(l_{12}^2 - 2l_0(l_{12} - l_0) - l_0^2) = \frac{k}{2}(l_{12}^2 - 2l_0l_{12} + 2l_0^2 - l_0^2) = \frac{k}{2}(l_{12}^2 - 2l_0l_{12} + l_0^2)$$

$$E_p = \frac{k}{2}(l_{12} - l_0)^2$$

(g)

Given $\vec{F}_1 = -\vec{F}_2 = \frac{C}{|\vec{r}_{12}|^3} \vec{r}_{12}$ find E_p . Chosen method: $E_p = -\int_{\vec{r}_0}^{\vec{r}} \vec{F} \cdot d\vec{r}$. This force is like inverse square gravity between two objects.



Sketch of force

In this case, because we have two particles and therefore two positions that change, we need to sum the two changes in potential energy. There are also two initial and two final positions.

$$E_p = -\int_{\vec{r}_{10}, \vec{r}_{20}}^{\vec{r}_1, \vec{r}_2} \vec{F}_1 \cdot d\vec{r}_1 + \vec{F}_2 \cdot d\vec{r}_2 = \int_{\vec{r}_{10}, \vec{r}_{20}}^{\vec{r}_1, \vec{r}_2} -\vec{F}_1 \cdot d\vec{r}_1 + \vec{F}_1 \cdot d\vec{r}_2 = \int_{\vec{r}_{10}, \vec{r}_{20}}^{\vec{r}_1, \vec{r}_2} \vec{F}_1 \cdot (d\vec{r}_2 - d\vec{r}_1)$$

Because $\vec{r}_1$ and $\vec{r}_2$ are two independent vectors, $d\vec{r}_2 - d\vec{r}_1 = d(\vec{r}_2 - \vec{r}_1) = d\vec{r}_{12}$. Therefore

$$\int_{\vec{r}_{10}, \vec{r}_{20}}^{\vec{r}_1, \vec{r}_2} \vec{F}_1 \cdot (d\vec{r}_2 - d\vec{r}_1) = \int_{\vec{r}_{10}, \vec{r}_{20}}^{\vec{r}_1, \vec{r}_2} \vec{F}_1 \cdot d\vec{r}_{12} = \int_{\vec{r}_{120}}^{\vec{r}_{12}} \frac{C}{|\vec{r}_{12}|^3} \vec{r}_{12} \cdot d\vec{r}_{12}$$

Using the fact that $\vec{A} \cdot d\vec{A} = |\vec{A}| d|\vec{A}|$, and defining $l_{12} = |\vec{r}_{12}|$, this is equal to

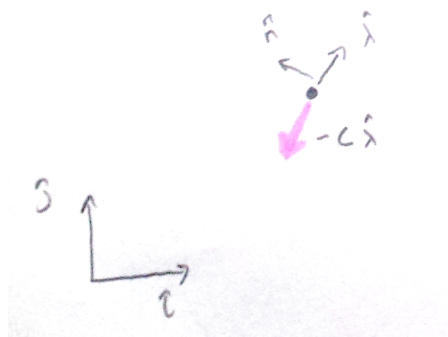
$$\int_{l_{120}}^{l_{12}} \frac{C}{l_{12}^2} dl_{12} = \frac{-C}{l_{12}} \Big|_{l_{120}}^{l_{12}} = C \left(\frac{1}{l_{120}} - \frac{1}{l_{12}} \right)$$

Choose $l_{120} = \infty$, to make the first fraction equal 0 (as an aside, gravitational potential energy is often defined as 0 at infinite distance, and defined as becoming more negative the closer you get to the object. Gravity is also an inverse square attraction!). Then

$$E_p = \frac{-C}{l_{12}}$$

(h)

Given $\vec{F} = -C\hat{\lambda}$ find E_p . Chosen method: $E_p = -\int_{\vec{r}_0}^{\vec{r}} \vec{F} \cdot d\vec{r}$. This force is a constant force in a random direction. It is like near-earth gravity (replace C with mg and $\hat{\lambda}$ with $\hat{j}$) but more general.



$$E_p = -\int_{\vec{r}_0}^{\vec{r}} -C\hat{\lambda} \cdot d\vec{r} = C\hat{\lambda} \cdot \left(\int_{\vec{r}_0}^{\vec{r}} d\vec{r} \right) \quad \text{Sketch of force}$$

Because $\hat{\lambda}$ is a constant vector, it is possible to pull it outside of the integral and dot it with the whole integral. This integral is also path independent, since it will evaluate to the change in position, which does not depend on the path.

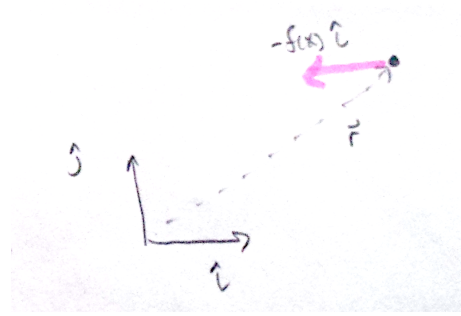
$$C\hat{\lambda} \cdot \left(\int_{\vec{r}_0}^{\vec{r}} d\vec{r} \right) = C\hat{\lambda} \cdot (\vec{r} - \vec{r}_0)$$

Choose $\vec{r}_0 = \vec{0}$ to find that

$$E_p = C\hat{\lambda} \cdot \vec{r}$$

(j)

Given $\vec{F} = -f(x)\hat{i}$ find E_p . Chosen method: $E_p = -\int_{\vec{r}_0}^{\vec{r}} \vec{F} \cdot d\vec{r}$. This force is an unknown force along an axis.



$$E_p = -\int_{\vec{r}_0}^{\vec{r}} -f(x)\hat{i} \cdot d\vec{r} = \int_{\vec{r}_0}^{\vec{r}} f(x)\hat{i} \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) = \int_{x_0}^x f(x)dx$$

Sketch of force

This integral is path independent.

Since we don't know the exact form of f , we can't simplify this integral further. However, it is important to choose a value of x_0 that will not cause the integral to become divergent. For instance, if $f(x) = \frac{1}{x-1}$, and we chose $x_0 = 1$, then $f(x_0)$ would be undefined and we couldn't solve it. The same problem would occur if $x_0 < 1$ and $x > 1$, because the integral would still include the point $f(1)$. Therefore,

$$E_p = \int_{x_0}^x f(x)dx \text{ with } x_0 \text{ chosen so the integral is not divergent}$$

Also see part (e) for a similar unknown force in the $\hat{e}_r$ direction.